3a) This choice basically fails because there is a valid decomposition that pumps both copies in the same way and keeps the string in the form xx, like for example, let v be the last a before the middle b, let $w = b$ and let x be the first a after that b, so pumping gives us $a^{k+1}ba^{k+1}b$, which is still of the form yy, so this string does not force a contradiction.

b) Let us assume for contradiction that L_6 is context-free, then by context-free pumping lemma, there exists a pumping length k such that every string $s \in L_6$ with $|s| \geq k$ can be written as $s = uvwxy$ with $|uvw| \leq k$, $|vx| > 0$ and $uv^i wx^i y \in L_6$ for all $i \geq 0$. Let us choose $s = a^k b^k a^k b^k$, since $s = (a^k b^k)(a^k b^k)$, so we have $s \in L_6$.

Now $|vwx| \leq k$, so the substring vwx can lie in at most one block or cross at most one boundary among the 4 blocks $a^k b^k a^k b^k$.

Now, let us consider some cases:

Case 1: vwx lies entirely in the first half $a^k b^k$

Then pumping changes only the first copy of $a^k b^k$, while the second copy remains unchanged.

So, for $i = 0$ or $i = 2$, the resulting string is no longer of the form xx. Hence $uv^iwx^iy \notin L$.

Case 2: vwx lies entirely in the second half $a^k b^k$

Symmetrically, pumping changes only the second copy leaving the first unchanged. Again, for $i = 0$ or $i = 2$, the resulting string is not of the form xx.

Case 3: vwx crosses the middle boundary between the 2 copies.

Then vwx contains some b's from the first half and some a's from the second half. Pumping changes the end of the first half and the beginning of the second half. So, after pumping, the first half and second half, are no longer identical: one side has a changed number of trailing b's and the other has a changed number leading a's. Thus the resulting is not of the form xx.

Hence, in every case, there exists i such that $uv^iwx^iy \notin L_6$, which contradicts the pumping lemma. Therefore, L_6 is not context-free.

3 ii) Let us assume for contradiction that L_7 is context-free pumping lemma, there exists a pumping length K such that every string $s \in L_7$ with $|s| \geq K$, which can be written as $s = uvwxy$ with $|vwx| \leq K$, $|vx| > 0$, $uv^iwx^iy \in L_7$ for all $i \geq 0$.

Let us choose that $s = a^K b^{K^2}$, which means that $s \in L_7$

Because $|vwx| \leq K$, the substring vwx lies either:

entirely in the a's, entirely in the b's or crosses the boundary between a's and b's. Now, let us consider cases.

Case 1: v and x contain only a's

let $t = |v| + |x| > 0$. Pumping down with $i = 0$ gives $uv^0wx^0y = a^{k-t}b^{k^2}$.

If this were in L_7, then we would need $k^2 = (k-t)^2$. But $t > 0$, so $(k-t)^2 < k^2$, impossible. Hence the pumped string is not in L_7.

Case 2: v and x contain only b's.

let $t = |v| + |x| > 0$. Pumping down with $i = 0$ gives $uv^0wx^0y = a^k b^{k^2 - t}$.

The number of a's is still k, so if this were in L_7, then number of b's would have to be exactly k^2.

But it is $K^2 - t \neq K^2$, which is a contradiction.

Case 3: vwx crosses the boundary between the a's and b's

Let t be the number of a's removed from vx and let s be the number of b's removed from vx when we pump down to $i=0$. Then $t>0$, $s \geq 0$ and since $|vwx| \leq K$, we have $s \leq K$. After pumping down, we get $a^{K-t} b^{K^2-s}$, if this were in L_7, then we would need $K^2 - s = (K-t)^2$, so $s = K^2 - (K-t)^2 = 2Kt - t^2 = t(2K-t)$, since $1 \leq t \leq K$, we have $s = t(2K-t) \geq 2K-1 > K$, which basically contradicts $s \leq K$, so this pumped string is also not in L_7. Since every case is gives us a contradiction, the pumping lemma fails, thus L_7 is not context-free.

4. (10 points) Use the CYK algorithm (the dynamic programming algorithm presented in class) to determine whether the context-free grammar below generates the string *aabaa*. Please **print this page** and fill in the provided table, and include that with the rest of your answers when scanning your handwritten work.

$$
\begin{aligned}
S &\rightarrow TA \mid BA \mid AB \mid b \\
A &\rightarrow AC \mid a \\
T &\rightarrow AB \\
B &\rightarrow b \\
C &\rightarrow a
\end{aligned}
$$